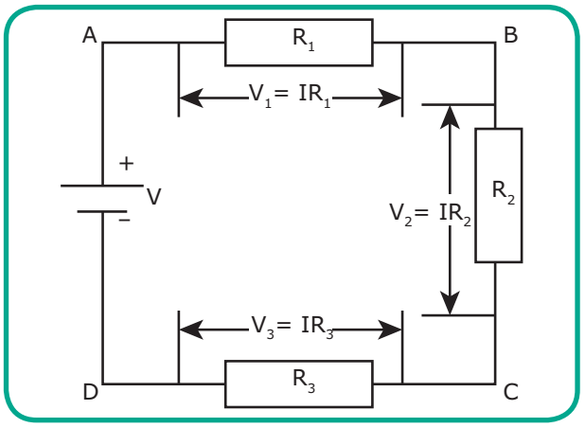


Figure 1.47 Kirchhoff's voltage law

In a closed circuit, the sum of the voltage drops is equal to the sum of the potential applied. This is called Kirchhoff's voltage law (shortly KVL)

i.e., $V = IR_1 + IR_2 + IR_3$

As per ohm's law

$$V = V_1 + V_2 + V_3$$

Example 1

Find the current flow through each resistor in the circuit shown in Figure 1.48, using Kirchhoff's law.

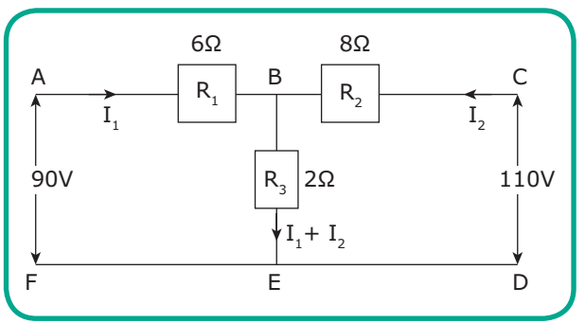


Figure 1.48

Solution

According to Kirchhoff's current law, the direction of current flow towards the junction is noted. According to voltage law, write down the KVL equation in the closed circuits.

1. ABEFA forms a closed circuit

$$6I_1+2(I_1+I_2) = 90$$

$$8I_1 + 2I_2 = 90 \qquad \text{Eqn.. 1.1}$$

2. CBEDC forms another closed circuit

$$8I_2+2(I_1+I_2) = 110$$

$$2I_1 + 10I_2 = 110 \qquad \text{Eqn.. } 1.2$$

To solve this, multiply Equation 1.2 by 4 on both sides, Therefore,

$$8I_1+40I_2 = 440 \qquad \text{Eqn.. } 1.3$$

Subtract Eq.1.3 from Eqn.1.1.

$$8I_1 + 40I_2 = 440$$
$$8I_1 + 2I_2 = 90 \ (-)$$
$$38I_2 = 350$$
$$I_2 = \frac{350}{38}$$
$$I_2 = 9.21\ A$$

By putting the value of I_2 in Eq.1.1, we can get the value of I_1

$$8I_1 + (2\ X\ 9.21) = 90$$
$$8I_1 = 90 - 18.42$$
$$I_1 = \frac{71.58}{8}$$
$$I_1 = 8.95\ A$$

From this, we found that

1. The current flows through resistor 6Ω is 8.95 A.
2. The current flows through 8Ω resistor is 9.21 A.
3. The current flows through 2Ω resistor is 8.95 + 9.21 = 18.16 A.

Example 2

By using Kirchhoff's law, calculate the current flowing through each resistor shown in the circuit diagram (Figure 1.49).

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